

ENGINEERING SPECIFICATION & DRYING MANUAL

Produce Processing & Dryer Compatibility Guide

Complete Dehydration Parameters, Moisture Reduction Standards, Optimal Temperatures, Batch Cycle Times, and SS304 Chamber Compatibility for 28+ Industrial Produce Varieties.

CALOR MEGA BATCH SYSTEMS | SS304 FOOD GRADE | PID TEMPERATURE CONTROL | ISO COMPLIANT

1. SYSTEM OVERVIEW & DEHYDRATION ARCHITECTURE

The CALOR MEGA industrial batch drying system is specifically engineered for multi-crop commercial processing. Equipped with 360C dynamic laminar cross-flow aerodynamics, Pt100 RTD precision thermal probes (±1CC accuracy), and automated electric moisture evacuation actuators, it guarantees consistent moisture reduction without burning, caramelization, case hardening, or nutrient degradation.

THERMAL RANGE

Ambient to 90CC

PID Digital ±1CC accuracy

AIRFLOW SYSTEM

360C Cross-Flow

Laminar axial dynamically balanced

CHAMBER & TRAYS

Food-Grade SS304

Optional SS316 high-acid proofing

2. CATEGORY-WISE DRYER COMPATIBILITY & PROCESSING MATRIX

CATEGORY: FRUITS

Crop / Produce	Moisture (In -> Out)	Temp & Batch Time	Shelf Life Extension	Dryer Compatibility & Output
Jackfruit <i>Tray: SS304 Perforated</i>	75% -> 5% Loss: 70% extracted	55CC - 62CC Time: 12 - 16 Hrs	3 Days -> 12+ Months Zero chemical preservatives	100% Compatible [Highest Yield] Output: Crispy Chips, Gluten-Free Flour
Banana <i>Tray: SS304 Wire Mesh</i>	70% -> 4% Loss: 66% extracted	58CC - 65CC Time: 10 - 14 Hrs	5 Days -> 12+ Months Zero chemical preservatives	100% Compatible [Export Grade] Output: Baked Chips, Nutritional Powder
Mango <i>Tray: SS304 Fine Mesh</i>	80% -> 6% Loss: 74% extracted	52CC - 60CC Time: 12 - 16 Hrs	4 Days -> 10+ Months Zero chemical preservatives	100% Compatible [High Margin] Output: Dried Slices, Amchur Powder
Pineapple <i>Tray: SS304 / SS316</i>	84% -> 7% Loss: 77% extracted	55CC - 60CC Time: 14 - 18 Hrs	3 Days -> 9+ Months Zero chemical preservatives	100% Compatible [Enzyme Preserved] Output: Dried Rings, Candied Tidbits
Citrus & Lemon <i>Tray: SS316 Acid-Proof</i>	85% -> 6% Loss: 79% extracted	48CC - 55CC Time: 10 - 14 Hrs	5 Days -> 12+ Months Zero chemical preservatives	100% Compatible [Color Stable] Output: Cocktail Wheels, Orange Ribbons
Papaya & Tutti-Frutti <i>Tray: SS304 Perforated</i>	88% -> 8% Loss: 80% extracted	55CC - 62CC Time: 12 - 15 Hrs	4 Days -> 12+ Months Zero chemical preservatives	100% Compatible [Bulk Commercial] Output: Bakery Cubes, Dried Strips

CATEGORY: VEGETABLES

Crop / Produce	Moisture (In -> Out)	Temp & Batch Time	Shelf Life Extension	Dryer Compatibility & Output
Mushroom (Oyster/Button) <i>Tray: SS304 Fine Mesh</i>	90% -> 6% Loss: 84% extracted	45CC - 52CC Time: 8 - 12 Hrs	2 Days -> 12+ Months Zero chemical preservatives	100% Compatible [Zero Discoloration] Output: Dried Caps, Umami Powder
Seasonal Veg (Carrots/Tomatoes) <i>Tray: SS304 Wire Mesh</i>	92% -> 8% Loss: 82% extracted	55CC - 65CC Time: 10 - 14 Hrs	4 Days -> 12+ Months Zero chemical preservatives	100% Compatible [Nutrient Intact] Output: Carrot Cubes, Sundried Tomatoes
Onion & Garlic Flakes <i>Tray: SS304 Fine Mesh</i>	84% -> 4% Loss: 80% extracted	50CC - 58CC Time: 8 - 12 Hrs	1 Week -> 18+ Months Zero chemical preservatives	100% Compatible [Industrial High Demand] Output: Kibbled Flakes, Garlic Granules

CATEGORY: SPICES AND HERBS				
Crop / Produce	Moisture (In -> Out)	Temp & Batch Time	Shelf Life Extension	Dryer Compatibility & Output
Turmeric & Ginger <i>Tray: SS304 Heavy Mesh</i>	80% -> 7% Loss: 73% extracted	55CC - 65CC Time: 16 - 22 Hrs	7 Days -> 24+ Months Zero chemical preservatives	100% Compatible [98% Curcumin Retained] Output: Polished Fingers, Sonth Ginger
Chili & Paprika <i>Tray: SS304 Perforated</i>	75% -> 8% Loss: 67% extracted	55CC - 60CC Time: 14 - 18 Hrs	6 Days -> 18+ Months Zero chemical preservatives	100% Compatible [Anti-Aflatoxin] Output: Whole Stemless, Chili Flakes
Cardamom & Pepper <i>Tray: SS304 Micro Mesh</i>	65% -> 8% Loss: 57% extracted	45CC - 50CC Time: 18 - 24 Hrs	5 Days -> 24+ Months Zero chemical preservatives	100% Compatible [Aroma Sealed] Output: Green Pods, Black Pepper
Culinary Herbs (Mint/Basil) <i>Tray: SS304 Micro Mesh</i>	75% -> 5% Loss: 80% extracted	40CC - 48CC Time: 6 - 9 Hrs	2 Days -> 12+ Months Zero chemical preservatives	100% Compatible [Zero Chlorophyll Loss] Output: Herb Flakes, Tea Cuts

CATEGORY: PLANTATIONS				
Crop / Produce	Moisture (In -> Out)	Temp & Batch Time	Shelf Life Extension	Dryer Compatibility & Output
Tea & Moringa <i>Tray: SS304 Micro Mesh</i>	78% -> 4% Loss: 74% extracted	45CC - 55CC Time: 6 - 8 Hrs	2 Days -> 18+ Months Zero chemical preservatives	100% Compatible [Superfood Certified] Output: Loose Leaf Tea, Moringa Powder
Coffee Beans <i>Tray: SS304 Perforated</i>	55% -> 11% Loss: 44% extracted	45CC - 52CC Time: 18 - 26 Hrs	1 Month -> 18+ Months Zero chemical preservatives	100% Compatible [Cup Score Maximized] Output: Parchment Coffee, Natural Cherries
Coconut & Copra <i>Tray: SS304 Wire Mesh</i>	50% -> 3% Loss: 47% extracted	60CC - 70CC Time: 10 - 14 Hrs	7 Days -> 12+ Months Zero chemical preservatives	100% Compatible [Low FFA Pure White] Output: Desiccated Flakes, Milling Copra

CATEGORY: GRAINS AND PULSES				
Crop / Produce	Moisture (In -> Out)	Temp & Batch Time	Shelf Life Extension	Dryer Compatibility & Output
Paddy & Rice <i>Tray: SS304 Micro Mesh</i>	24% -> 12% Loss: 12% extracted	42CC - 48CC Time: 6 - 10 Hrs	1 Month -> 24+ Months Zero chemical preservatives	100% Compatible [Zero Grain Breakage] Output: Parboiled Rice, Basmati Grain
Corn & Maize <i>Tray: SS304 Perforated</i>	28% -> 13% Loss: 15% extracted	50CC - 58CC Time: 8 - 12 Hrs	3 Weeks -> 18+ Months Zero chemical preservatives	100% Compatible [Mold Free Guarantee] Output: Dehydrated Sweet Corn, Feed Corn
Millet & Sorghum (Ragi) <i>Tray: SS304 Micro Mesh</i>	20% -> 10% Loss: 10% extracted	45CC - 50CC Time: 5 - 8 Hrs	2 Months -> 24+ Months Zero chemical preservatives	100% Compatible [High Germination] Output: Cleaned Ragi, Sorghum Grain
Pulses & Lentils (Dal) <i>Tray: SS304 Micro Mesh</i>	22% -> 9% Loss: 13% extracted	45CC - 52CC Time: 6 - 9 Hrs	1 Month -> 24+ Months Zero chemical preservatives	100% Compatible [Pest Immune Storage] Output: Split Dal, Whole Moong
Soybeans & Legumes <i>Tray: SS304 Wire Mesh</i>	22% -> 10% Loss: 12% extracted	45CC - 50CC Time: 6 - 8 Hrs	1 Month -> 18+ Months Zero chemical preservatives	100% Compatible [Zero Protein Loss] Output: Soy Flour Base, Edible Legumes

CATEGORY: NUTS AND TUBERS				
Crop / Produce	Moisture (In -> Out)	Temp & Batch Time	Shelf Life Extension	Dryer Compatibility & Output
Cashew & Kernels <i>Tray: SS304 Fine Wire</i>	16% -> 4% Loss: 12% extracted	65CC - 72CC Time: 6 - 8 Hrs	1 Month -> 18+ Months Zero chemical preservatives	100% Compatible [Premium White W180-W320] Output: Whole White Kernels, Roasted Cashews
Areca Nut (Supari) <i>Tray: SS304 Perforated</i>	60% -> 10% Loss: 50% extracted	58CC - 65CC Time: 24 - 36 Hrs	5 Days -> 24+ Months Zero chemical preservatives	100% Compatible [Glossy Anti-Fungal] Output: Chali Whole, Boiled Tukra
Cassava & Tapioca <i>Tray: SS304 Wire Mesh</i>	65% -> 9% Loss: 56% extracted	55CC - 65CC Time: 10 - 14 Hrs	2 Days -> 12+ Months Zero chemical preservatives	100% Compatible [Cyanide-Free Quick Dry] Output: Tapioca Chips, High-Grade Flour
Peanuts & Oilseeds <i>Tray: SS304 Fine Mesh</i>	30% -> 6% Loss: 24% extracted	45CC - 52CC Time: 6 - 10 Hrs	2 Weeks -> 12+ Months Zero chemical preservatives	100% Compatible [High Oil Quality] Output: Snack Peanuts, Hulled Sesame

CATEGORY: SPECIALTY & BOTANICALS

Crop / Produce	Moisture (In -> Out)	Temp & Batch Time	Shelf Life Extension	Dryer Compatibility & Output
Medicinal Plants (Ashwagandha) <i>Tray: SS304 Wire Mesh</i>	200% to 5% Loss: 77% extracted	40CC - 46CC Time: 8 - 12 Hrs	3 Days -> 24+ Months Zero chemical preservatives	100% Compatible [Pharma Alkaloid Retained] Output: Dried Roots, Botanical Extracts
Flowers & Petals (Rose/Marigold) <i>Tray: SS304 Silk Mesh</i>	100% to 6% Loss: 74% extracted	38CC - 45CC Time: 6 - 10 Hrs	2 Days -> 12+ Months Zero chemical preservatives	100% Compatible [Pigment & Shape Retained] Output: Lutein Petals, Culinary Rose Petals
Fish & Seafood (Dry Fish/Shrimp) <i>Tray: SS316 Washable Mesh</i>	200% to 10% Loss: 58% extracted	48CC - 56CC Time: 12 - 16 Hrs	1 Day -> 8+ Months Zero chemical preservatives	100% Compatible [Zero Fly/Dust Contamination] Output: Hygienic Dry Fish, Dried Shrimp

3. DRYER OPERATION & TRAY LOADING PROTOCOLS

1. Tray Loading Density Guidelines

For thin leafy crops (tea, moringa, herbs), maintain 2.0 to 3.0 kg/m² to prevent clumping. For sliced fruits (jackfruit, banana, mango), arrange single-layer slices at 4.5 to 6.0 kg/m². Dense grains and seeds can be loaded up to 8.0 kg/m² with periodic agitation.

2. Multi-Stage Temperature Profiling

High initial moisture crops benefit from a 2-stage heating protocol: Start at 60CC - 65CC for the first 2-3 hours with dampers at 80% open to rapidly evacuate free surface water, then reduce temperature to 50CC - 55CC with dampers at 30% for deep cellular moisture removal without case hardening.

3. Food Safety & Hygiene Standards

CALOR MEGA chambers are fabricated from electropolished SS304/SS316. Removable tray racks are fully dishwasher safe and compatible with steam-cleaning CIP (Clean-In-Place) procedures, adhering to international USFDA 21 CFR and European EC 1935/2004 food contact directives.

4. Energy Conservation & ROI Optimization

Double-walled rockwool insulation (75mm high density) ensures 99% thermal energy retention, keeping casing exterior within 3CC of ambient temperature. Continuous operation reduces drying electricity cost by up to 42% compared to conventional open electric batch ovens.

NEED CUSTOM DRYING PARAMETERS OR TESTING FOR UNLISTED CROPS?

Our engineering and food science application laboratory provides pilot batch trial runs and customized temperature-airflow recipes.

Website: <https://sreyastejas777-png.github.io/Technology-Calor-Mega/> | **WhatsApp Technical Desk:** +91 99999 99999